

Booking Patterns for Cancellation Risk Management

A Reproducible Unsupervised Clustering Study of Hotel Booking Demand

Unsupervised Learning — 2025/2026 · NOVA FCT

Repository tag:

<https://github.com/AfonsoCSimoes/UnsupervisedLearning-Project.git>

June 2026

Afonso Simões (73204) José Moutinho (73129) Luís Nunes (73216)

Turno: TP1 Professor: Susana Nascimento

Abstract

Hotel booking cancellations impose severe operational costs, representing 37.0% of all bookings in this study’s dataset. We conduct a fully reproducible clustering study to discover structurally distinct *pre-arrival* booking profiles using exclusively features available at the time of reservation. Our research question is: “*Do bookings made far in advance through indirect channels naturally emerge as a distinct cluster, and how does this profile associate with observed cancellation rates?*” We construct the **R-EUCLID-standard-countryTop15-noADR** representation (55-dimensional Euclidean matrix), apply K-Means and Gaussian Mixture Models (GMM) as complementary clustering families, and employ iK-Means for data-driven K estimation, evaluating across 5 random seeds and $K \in \{3, 4, 5, 6, 7, 8, 11, 14\}$. $K = 5$ is selected as a principled compromise between internal validity, bootstrap stability and cluster interpretability. Post-hoc profiling reveals five structurally distinct profiles with cancellation rates spanning from 5.5% to 98.9%. Critically, Cluster 2 ($n = 5,886$; 4.9% of bookings) exhibits features consistent with speculative advance bookings: lead times 104.6% above the global mean, a 1,688% excess of prior cancellations, and a 404% excess of non-refundable deposits, arriving almost entirely via indirect channels. These findings support targeted countermeasures rather than blanket cancellation policies.

Keywords: hotel booking demand; cancellation risk; clustering; K-Means; GMM; iK-Means; mixed-type data; reproducibility.

Contents

1	Introduction	3
2	Dataset and Preprocessing	4
2.1	Dataset Documentation	4
2.2	Data Quality and Feature Governance	4
2.3	The R-EUCLID Representation	5
3	Clustering Methods	6
3.1	K-Means (Baseline)	6
3.2	iK-Means (Data-Driven K Estimation)	7
3.3	Gaussian Mixture Models (Alternative Family)	7
4	Experimental Protocol	7
5	Results and Cluster Interpretation	8
5.1	Data Preparation Outcomes	8
5.2	K Selection and Model Comparison	8
5.3	Robustness Analysis	10
5.4	Cluster Profiles and Interpretation	11
6	Discussion	13
7	Limitations, Ethics, and Reproducibility	14
8	Conclusion	15

1 Introduction

Hotel revenue management depends on accurate demand forecasting. A booking that is ultimately cancelled generates no revenue yet consumes inventory and may displace a genuine guest. With a global cancellation rate of 37.0% in the course-release dataset, understanding *which types of bookings* are structurally prone to cancellation and why, is a decision-relevant problem.

Supervised classification of cancellation risk requires labelled historical data and assumes that past cancellation outcomes predict future individual behaviour. Unsupervised clustering, by contrast, operates on **pre-arrival behavioural signals** (booking channel, lead time, stay design, prior customer history) and groups bookings by *inherent structural similarity*, without being told which ones eventually cancelled. This allows the discovery of natural booking profiles that the hotel can act upon at reservation time—before any outcome is known.

Research Questions

RQ1 (Framing). Do bookings made far in advance through indirect channels naturally emerge as a structurally distinct pre-arrival cluster, and how does this profile associate with observed cancellation rates?

RQ2 (Representation). Which preprocessing and encoding choices produce a principled, leakage-safe Euclidean representation of mixed booking data, and how do those choices influence perceived similarity?

RQ3 (Modelling). Which clustering family and configuration yields the most coherent and stable solution under the chosen evaluation protocol?

RQ4 (Interpretation). What do the discovered clusters represent, and do they provide actionable information for cancellation risk management?

RQ5 (Robustness). How sensitive are the conclusions to reasonable variations in pre-processing scaler and the inclusion or exclusion of the `hotel` feature?

Contribution

This study delivers: (i) a fully reproducible, leakage-safe preprocessing pipeline for mixed tabular data; (ii) a principled comparison of K-Means, GMM, and iK-Means on the same representation; (iii) bootstrap and seed-based stability evidence; (iv) controlled StandardScaler vs RobustScaler robustness checks; and (v) post-hoc profiling of five interpretable booking segments linked directly to the cancellation-risk question.

2 Dataset and Preprocessing

2.1 Dataset Documentation

Table 1: Dataset documentation and governance record.

Item	Description
Dataset & version	Hotel Booking Demand, course release v1.
Unit of analysis	One row = one hotel booking record (arrived or cancelled).
Size & time span	119,388 bookings; 32 attributes; arrivals 1 Jul 2015 to 31 Aug 2017.
Hotels	Two properties: Resort Hotel (Algarve, 40,060 bookings) and City Hotel (Lisbon, 79,330 bookings).
Source & reference	António et al. [2019] ; CC BY 4.0.
SHA-256 hash	7c2ae42a7353905ea136e5c2287f17c92c5435826598bfbb8491c6f0c7b1fc06
Raw-data handling	Raw CSV not committed to version control. Instructions for obtaining and verifying the file are provided in <code>README.md</code> . The full pipeline can be reproduced via <code>run_all.py</code> .

2.2 Data Quality and Feature Governance

Segmentation (index) time. We segment bookings *immediately after the initial reservation is recorded*. All clustering features must be available and fixed at that moment. Post-arrival and post-decision variables are excluded from clustering inputs and reserved exclusively for post-hoc profiling.

Missingness and outliers. The `children` field has a small fraction of missing values, imputed as zero (no children recorded at booking time, consistent with the dataset description in [António et al. \[2019\]](#)). The `country` field is missing for 487 records; these rows are assigned to the “Other” group. Remaining categorical nulls are filled with “Unknown”. ADR presents two extreme cases: 4 records with negative values and a small number of entries exceeding €5,000 (physically implausible). Because ADR is excluded from the clustering matrix (see below), these rows are removed as a *data-quality* step prior to all analysis ($\text{adr} \geq 0$ and $\leq 5,000$), reducing the working set to 119,388 bookings.

Leakage control. Table 2 lists all features and their governance roles. The following fields are **excluded from clustering inputs**:

- `is_canceled`, `reservation_status`, `reservation_status_date`: outcome and post-event fields; realised after the booking is complete.
- `agent`, `company`: raw high-cardinality identifiers (333 and 352 distinct values); excluded to prevent overfitting and distance domination.
- `assigned_room_type`, `booking_changes`: post-decision, realised after the index time.
- `arrival_date_year`: excluded because it conflates temporal drift with behavioural similarity, and the dataset spans only two full years.
- `adr`: excluded from the clustering matrix to avoid price domination and to preserve a *behavioural* (not commercial-value) segmentation. ADR is retained exclusively for post-hoc profiling.

Table 2: Feature governance table (selected variables).

Variable	Role	Transformation	Rationale
lead_time	Input	log1p $\rightarrow$ scale	Right-skewed (0–737 days); log-compression reduces extreme-value dominance in Euclidean distance.
previous_cancellations	Input	log1p $\rightarrow$ scale	Heavy right tail; directly captures cancellation history.
previous_bookings_not_canceled	Input	log1p $\rightarrow$ scale	Complement of cancellation history; signals reliability.
total_nights	Input (derived)	scale	stays_in_weekend_nights + stays_in_week_nights ; single stay-length feature.
party_size	Input (derived)	scale	adults + children + babies ; single party-composition feature.
distribution_channel	Input	full OHE	Essential for RQ1 (channel strategy).
market_segment	Input	full OHE	Captures booking route and customer class.
deposit_type	Input	full OHE	No Deposit / Non Refund / Refundable signals commitment level.
customer_type	Input	full OHE	Transient / Contract / Group classification.
country	Input	top-15 + Other, OHE	177 distinct values; top-15 countries cover the vast majority.
arrival_date_month	Input	full OHE	Captures seasonality without imposing ordinal distance.
hotel	Input	OHE	Main representation includes hotel type; noHotel variant tested.
is_canceled	Profile only	—	Outcome variable; used exclusively post-hoc.
adr	Profile only	—	Price proxy; excluded from behavioural clustering.
agent, company	Excluded	—	High-cardinality identifiers; no meaningful behavioural signal.
reservation_status_date	Excluded	—	Post-event; realised after cancellation/check-out.

2.3 The R-EUCLID Representation

We define the main representation as **R-EUCLID-standard-countryTop15-noADR-withHotel**. After applying the transformations in Table 2, each booking becomes a 55-dimensional vector:

$$\mathbf{x}_i = \left[\underbrace{z_1, \dots, z_5}_{5 \text{ scaled numerical}}, \underbrace{o_1, \dots, o_{50}}_{50 \text{ one-hot indicators}} \right] \in \mathbb{R}^{55} \quad (1)$$

Similarity is measured by ordinary Euclidean distance:

$$d^2(i, \ell) = \|\mathbf{x}_i - \mathbf{x}_\ell\|_2^2 = \sum_{j=1}^5 (z_{ij} - z_{\ell j})^2 + \sum_{q=1}^Q \sum_{r=1}^{m_q} (o_{iqr} - o_{\ell qr})^2 \quad (2)$$

where z_{ij} is the scaled value of numerical variable j , and $o_{iqr} \in \{0, 1\}$ is the one-hot indicator for category r of categorical variable q (with $Q = 7$ categorical fields and m_q the number of retained categories for field q).

Why log1p for skewed numericals? `lead_time` ranges from 0 to 737 days; the squared-distance contribution of a raw difference of 700 days would be 490,000, dominating the categorical block. Log-compression reduces this to a manageable scaled contribution while preserving ordering.

Why full one-hot without rescaling? One-hot columns are binary by construction (0 or 1). Applying `StandardScaler` to a rare category (e.g., `deposit_type = Refundable`, which occurs in $< 1\%$ of rows) would artificially amplify its scaled value, making that rare event dominate distances. Keeping OHE columns at $\{0, 1\}$ preserves the intended distance property: any mismatch on a nominal category contributes exactly 2 to $d^2(i, \ell)$.

Why top-15 countries? 177 distinct country codes would produce a 177-column block, heavily sparse. The top 15 countries account for the overwhelming majority of bookings; all remaining are collapsed into “Other”.

Sensitivity checks. Two controlled variants are evaluated alongside the main representation: **RobustScaler** instead of `StandardScaler` on numerical variables only (tests whether heavy-tailed outliers dominate distances), and **noHotel** (removes the hotel binary feature to test whether City/Resort membership artificially dominates cluster separation).

3 Clustering Methods

All methods operate on the same R-EUCLID matrix (Equation 1) using Euclidean distance, ensuring that evaluation indices are computed in a consistent metric space.

3.1 K-Means (Baseline)

K-Means minimises the within-cluster sum of squared Euclidean distances by iteratively assigning each point to the nearest centroid and updating centroids as the within-cluster mean:

$$J = \sum_{k=1}^K \sum_{i \in S_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2 \quad (3)$$

K-Means is the natural baseline for an R-EUCLID representation: it operates directly in Euclidean space, scales well to 119,388 records, and its centroids provide interpretable within-cluster average profiles. We use `n_init=10` to reduce sensitivity to random initialisation.

Why K-Means is appropriate here: (i) the representation is Euclidean by design; (ii) the dataset is large enough that the assumption of reasonably convex clusters (required

for K-Means) is supported by the data; (iii) it is computationally fast (< 2 s per run at $K = 5$), allowing thorough grid search and stability analysis.

3.2 iK-Means (Data-Driven K Estimation)

iK-Means [Mirkin, 2012] is a deterministic initialisation procedure that identifies *anomalous clusters*, directions of maximum scatter from the global mean, and extracts them sequentially. It does not require a predefined K ; instead, it determines K automatically subject to a minimum cluster size (`min_cluster_size` = 100). For each identified anomalous pattern, a centroid in the standardised feature space is obtained, and these centroids are then used as seeds for a final K-Means run.

Why iK-Means is included: it provides a data-driven benchmark for K , free from the grid-search bias of the K-Means elbow analysis. Its K estimate is compared against the chosen $K = 5$ to assess whether the manual selection is consistent with the intrinsic structure of the data.

3.3 Gaussian Mixture Models (Alternative Family)

GMMs model the data as a weighted sum of K multivariate Gaussian components and assign soft memberships:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (4)$$

where $\boldsymbol{\Sigma}_k$ is a full covariance matrix fitted per component. GMMs differ from K-Means in three key ways: (i) they accommodate ellipsoidal, overlapping clusters rather than spherical ones; (ii) they produce probabilistic (soft) assignments, exposing genuinely ambiguous bookings; (iii) model selection via AIC/BIC provides an information-theoretic criterion independent of silhouette-type indices.

We use `covariance_type='full'`, `n_init=3`, and the same seed list as K-Means. K is selected by examining AIC and BIC curves across $K \in \{3, \dots, 8\}$ (Figure 6). Because both criteria decrease without a clear elbow through $K = 8$, and to enable a direct comparison with the K-Means solution on the same K , we fix $K = 5$ for the GMM as well, explicitly labelling this a practical and comparative choice rather than an information-theoretic optimum.

4 Experimental Protocol

Parameter ranges (pre-defined). The K grid $K \in \{3, 4, 5, 6, 7, 8, 11, 14\}$ is pre-specified before any model is run. Values below 3 lack practical interpretability; 14 is included because iK-Means suggests up to 14 anomalous patterns in the main representation (Table 4), providing an upper anchor. Unbounded search is not used.

Seeds and resampling. K-Means is run on 5 fixed seeds $\{42, 123, 456, 789, 999\}$ for every $(K, \text{representation})$ combination. GMM is similarly run on the same 5 seeds with `n_init=3` per run. All internal indices reported are means $\pm$ standard deviations across seeds.

Bootstrap stability (K-Means). For each K , we draw 20 bootstrap resamples (with replacement, same size as training set). For every pair of resamples (i, j) , we compute the Adjusted Rand Index (ARI) on the intersection of record indices present in both resamples. This avoids label-alignment issues across disjoint subsets. We report mean, standard deviation, minimum, and maximum pairwise ARI across all $\binom{20}{2} = 190$ pairs.

GMM seed stability. We report pairwise ARI between hard-assignment label vectors across the 5 seeds for each K .

Internal validity indices.

- **Silhouette score** (higher is better): measures how much more similar a booking is to its own cluster than to the nearest other cluster. Computed with a 30,000-point sample for efficiency.
- **Calinski-Harabasz (CH)** (higher is better): ratio of between-cluster to within-cluster dispersion.
- **Davies-Bouldin (DB)** (lower is better): average ratio of within-cluster scatter to between-cluster separation.

All indices are computed in the same R-EUCLID Euclidean space.

Experiment log. Every run is logged to `tables/experiments.csv` with columns: `representation_id`, `method`, `k`, `seed`, `silhouette`, `calinski_harabasz`, `davies_bouldin`, `runtime_seconds`, `sample_rule`, `parameters`, `mean_ARI`, `std_ARI`, `min_ARI`, `max_ARI`.

Robustness / sensitivity checks.

- R1. StandardScaler vs RobustScaler:** the numerical block (5 variables) is scaled with median/IQR instead of mean/std. One-hot columns remain 0/1 in both cases. This tests whether extreme values in `lead_time`, `previous_cancellations`, etc. drive cluster separation.
- R2. withHotel vs noHotel:** the `hotel` binary feature is added or removed. This tests whether the City/Resort split dominates the clustering irrespective of booking behaviour.

All four combinations $\{\text{standard, robust}\} \times \{\text{withHotel, noHotel}\}$ are run with the full K-grid and 5 seeds, producing 324 experiment rows in total.

5 Results and Cluster Interpretation

5.1 Data Preparation Outcomes

After applying the governance rules in Table 2 and cleaning, the analytical matrix has **119,388 rows and 55 columns** (5 scaled numerical + 50 one-hot indicators). The full feature list is saved to `tables/final_feature_space.csv`.

5.2 K Selection and Model Comparison

K-Means grid analysis. Table 3 reports all internal indices for the main representation (R-EUCLID-standard-countryTop15-noADR-withHotel), averaged across 5 seeds.

Table 3: K-Means evaluation grid (main representation, mean $\pm$ std across 5 seeds). Selected configuration highlighted.

K	Silhouette	CH	DB	ARI (mean)	ARI (std)	ARI (min)	ARI (max)
3	0.162 $\pm$ 0.018	15,246	1.894	0.876	0.130	0.710	0.998
4	0.147 $\pm$ 0.014	15,406	1.756	0.660	0.277	0.195	0.997
5	0.152$\pm$0.013	15,727	1.701	0.723	0.270	0.261	0.997
6	0.140 $\pm$ 0.001	15,379	1.760	0.984	0.006	0.968	0.994
7	0.139 $\pm$ 0.006	14,633	1.760	0.923	0.141	0.618	0.997
8	0.147 $\pm$ 0.012	13,913	1.752	0.976	0.053	0.814	0.998
11	0.145 $\pm$ 0.008	11,957	1.708	0.780	0.085	0.639	0.997
14	0.145 $\pm$ 0.004	10,657	1.662	0.704	0.085	0.512	0.906

Why $K = 5$? No single metric produces an unambiguous elbow (Figure 1), which is consistent with the overlapping, mixed-type structure of booking data. The selection of $K = 5$ is justified on three complementary grounds:

1. **CH is maximised at $K = 5$** (15,727) among all tested values, indicating optimal between-to-within cluster variance ratio.
2. **DB is minimised among $K \leq 8$ at $K = 5$** (1.701), indicating good within-cluster compactness relative to separation.
3. **Profile interpretability:** the 5-cluster solution isolates a compact but critical cluster (C2, $n = 5,886$) with a near-perfect cancellation rate (98.9%) and a set of highly distinct behavioural markers. This cluster would be merged into larger groups at $K < 5$, losing the specific risk signal. At $K > 5$, the additional clusters represent increasingly fine-grained subdivisions that do not add actionable resolution for cancellation risk management.

We emphasise that $K = 5$ should be understood as a *practical compromise* rather than a unique global optimum; the modest silhouette values (0.08–0.16 across the grid) reflect the genuine difficulty of partitioning a high-dimensional mixed-type space.

iK-Means results. Table 4 reports iK-Means outcomes across all representations.

Table 4: iK-Means results across representations (deterministic; no seed).

Representation	K (auto)	Silhouette	CH	DB	Runtime (s)
Standard, withHotel	14	0.139	10,029	1.836	69.8
Standard, noHotel	13	0.149	11,177	1.771	57.5
Robust, withHotel	11	0.132	9,510	2.027	57.4
Robust, noHotel	10	0.140	11,608	1.879	57.6

iK-Means identifies between 10 and 14 anomalous patterns depending on the representation, suggesting that the data supports more fine-grained structure than $K = 5$. However, at $K = 14$, the cluster profiles become difficult to distinguish and the actionable signal dilutes across numerous small groups. We use the iK-Means K estimate as a *ceiling* for the grid search and as confirmation that $K \in \{5, \dots, 14\}$ is a reasonable operating range.

Baseline vs GMM comparison. Table 5 compares all three methods at $K = 5$ on the main representation.

Table 5: Method comparison at $K = 5$, main representation (R-EUCLID-standard-countryTop15-noADR-withHotel). K-Means and iK-Means metrics are in the same Euclidean space; GMM silhouette is lower due to soft, overlapping components.

Method	K	Sil.	CH	DB	ARI (mean)	ARI (std)	Runtime
K-Means (std)	5	0.152	15,727	1.701	0.723	0.270	1.15 s
K-Means (robust)	5	0.151	15,474	1.956	0.992	0.003	0.83 s
iK-Means (std)	14	0.139	10,029	1.836	—	—	69.8 s
GMM full cov (std)	5	0.024	2,908	4.636	0.564	0.208	34.4 s

K-Means substantially outperforms GMM on all three Euclidean-based internal indices. This is expected: GMM with full covariance matrices models ellipsoidal, overlapping Gaussian components, and when evaluated in the Euclidean space used for clustering (not the GMM-native likelihood space), the hard Euclidean indices naturally favour the K-Means partition. The key contribution of GMM is complementary: it models cluster shape and produces soft memberships, acting as a robustness check on the cluster structure found by K-Means.

Despite lower internal indices, the GMM assignment certainty is remarkably high: as shown in Figure 7, the vast majority of bookings are assigned with probability ≈ 1.0 to their dominant component. This indicates that the data has a strong, near-hard cluster structure even when a probabilistic model is used, lending support to the K-Means partition.

The consensus heatmap (Figure 8) shows that K-Means Cluster 2 (the critical high-risk group) maps almost entirely to GMM Cluster 2, confirming that this high-cancellation segment is robustly identified regardless of the modelling family.

5.3 Robustness Analysis

StandardScaler vs RobustScaler. Figures 2a and 2b show bootstrap ARI profiles for both scalers. Both exhibit identical pattern: near-perfect ARI at $K = 6$ (0.984 ± 0.006), high ARI at $K = 3$ (0.876 ± 0.129), and moderate variance at $K = 4$ and $K = 5$. At $K = 5$ the ARI for the RobustScaler representation is 0.992 ± 0.003 (min 0.985, max 0.998), dramatically more stable than the StandardScaler result (0.723 ± 0.270 , min 0.261, max 0.997). This suggests that a small number of extreme values in `lead_time` and `previous_cancellations` can occasionally induce a different partition under StandardScaler, whereas RobustScaler is insensitive to those extremes. However, as shown in Table 5, the RobustScaler silhouette at $K = 5$ (0.151) is nearly identical to the StandardScaler result (0.152), and the post-hoc cluster profiles (cancellation rates, lead time distributions, channel profiles) remain consistent. We conclude that the cluster *interpretation* is robust across scalers, while the RobustScaler produces a more numerically stable solution.

withHotel vs noHotel. Removing the `hotel` binary feature reduces the feature space from 55 to 53 dimensions. K-Means without hotel achieves $CH = 16,452$ (vs 15,727 with hotel), suggesting a marginally better separation-to-compactness ratio without the binary City/Resort signal. iK-Means identifies $K = 13$ without hotel vs $K = 14$ with hotel, indicating that the hotel feature accounts for approximately one additional separable cluster pattern (likely the fine-grained City/Resort distinction). The post-hoc cancellation profiles remain consistent. We adopt the `withHotel` representation as primary because hotel type is known at booking time and is a practically meaningful variable for risk management.

5.4 Cluster Profiles and Interpretation

Figure 3 displays the profile heatmap showing relative deviation from the global mean for each of the 10 key profiling variables. Figures 5a and 5b show the lead time and ADR distributions per cluster. Figure 4 shows post-hoc cancellation rates per cluster. Figure 9 shows a PCA projection comparing K-Means and GMM assignments.

Table 6: Cluster profile summary (K-Means, $K = 5$, main representation). Cancellation rates are *post-hoc*: derived from `is_canceled` after cluster assignment. ADR is similarly a post-hoc profiling variable, not a clustering input.

Cluster	Size (n)	Size (%)	Cancel rate	Median ADR (€)	Main behavioural markers	Profile label
C0	2,040	1.7%	5.5%	55	Very short lead time (-85%); high prior cancellations ($+567\%$); predominantly direct channel ($+74\%$); zero non-refund deposits (-96%); small party (-38%).	Last-min. serial cancellers (reformed)
C1	18,327	15.4%	33.8%	100	Long stays ($+127\%$); near-zero prior cancellations (-97%); moderate lead time ($+28\%$); no non-refund (-86%).	Extended-stay reliable bookers
C2	5,886	4.9%	98.9%	72	Very long lead time ($+105\%$); extreme prior cancellations ($+1,688\%$); non-refundable deposits ($+405\%$); indirect channel (-79% direct).	Speculative high-risk bookings
C3	27,502	23.0%	14.0%	87	Very short lead time (-93%); zero prior cancellations (-100%); strongly direct ($+128\%$); zero non-refund (-92%); short stays.	Last-minute direct low-risk bookers
C4	65,633	55.0%	43.0%	100	Above-average lead time ($+24\%$); zero prior cancellations (-100%); indirect channel (-54% direct); elevated non-refund ($+30\%$).	Standard advance bookings (majority)

C0 — Last-Minute Serial Cancellers (Reformed). This is the smallest cluster ($n = 2,040$; 1.7%), characterised by a paradox: guests in this group have a historical cancellation rate 567% above the global mean, yet their observed cancellation rate in this dataset is only 5.5%—the lowest of all clusters. The reconciliation lies in the lead time: these guests book almost at the last minute (lead time -85% below global mean,

with a median of approximately 10 days), making cancellation physically unlikely since the arrival date is imminent. They book directly (+74% above global mean for direct channel), pay standard rates (ADR median $\approx$ €55), and carry no non-refundable deposit. The interpretation is that this group may represent guests who have historically used the system speculatively (booking far in advance and then cancelling), but whose *current* booking is opportunistic and committed (they would not book last-minute if they were not intending to stay).

C1 — Extended-Stay Reliable Bookers. At $n = 18,327$ (15.4%), this cluster is characterised by exceptionally long stays (+127% above mean total nights), near-zero prior cancellation history (−97%), and moderate advance planning (lead time +28%). The cancellation rate of 33.8% is below the global mean. ADR median is €100, consistent with longer leisure stays at resort hotels. These guests represent a valuable, reliable segment: high revenue contribution per booking, low cancellation risk, and demonstrated reliability.

C2 — Speculative High-Risk Bookings (Answer to RQ1). This cluster ($n = 5,886$; 4.9%) directly answers the research question. The combination of markers is consistent with a speculative booking profile:

- **Lead time:** median $\approx$ 250 days (more than 8 months before arrival); +105% above global mean.
- **Prior cancellations:** +1,688% above global mean. These guests are systematically serial cancellers.
- **Non-refundable deposit:** +405% above mean. A counterintuitive finding: these bookings carry non-refundable deposits, suggesting they are made through channels that require upfront commitment but that the guests (or intermediaries) still cancel, forfeiting the deposit.
- **Channel:** predominantly indirect (−79% direct), consistent with travel agents or online booking platforms that facilitate large-scale speculative reservations.
- **Observed cancellation rate:** 98.9%, effectively all bookings in this cluster are cancelled.

This cluster’s post-hoc cancellation rate of 98.9% should be interpreted cautiously: it reflects observed historical behaviour in this dataset, not a deterministic rule. The cluster was formed *without* using `is_canceled` as an input; the near-perfect cancellation alignment is an *emergent* discovery from purely pre-arrival behavioural signals.

C3 — Last-Minute Direct Low-Risk Bookers. With $n = 27,502$ (23.0%) and a cancellation rate of only 14.0%, this cluster is the most reliable among the larger segments. Key markers: booking is almost exclusively last-minute (lead time −93%, median $\approx$ 10 days), exclusively through direct channels (+128%), with zero prior cancellation history and zero non-refundable deposits. The short lead time mechanically reduces cancellation probability (the guest has committed close to arrival), and the direct channel filters out intermediary-driven speculative bookings.

C4 — Standard Advance Bookings (Majority Cluster). The largest cluster ($n = 65,633$; 55.0%) represents the mainstream hotel booker: advance planning (lead time +24%), no cancellation history, moderate use of indirect channels, and a somewhat elevated non-refundable deposit (+30%). The cancellation rate of 43.0% is above the global mean,

driven primarily by the combination of advance booking and indirect channels, but is diffuse and lacks the extreme structural markers of C2.

6 Discussion

Does the early/indirect profile naturally emerge? Yes. Cluster C2 demonstrates that bookings made far in advance through indirect channels do form a structurally distinct group, identifiable from pre-arrival features alone. This confirms the core hypothesis of RQ1. Critically, the group is defined by the *combination* of long lead time, high prior cancellations, and non-refundable deposit, not by any single feature. A univariate rule (e.g., “flag any booking with lead time > 200 days”) would incorrectly classify many C4 bookings (which also have above-average lead times but nearly zero prior cancellations).

How did preprocessing choices influence results? The R-EUCLID representation was designed to treat all behavioural signals symmetrically. The $\log_{1p}$ transformation of `previous_cancellations` is critical: without it, the extreme values in C2 (+1,688% above mean, corresponding to guests with many prior cancellations) would dominate the Euclidean distance and likely force all high-cancellation-history records into a single large cluster regardless of their other behavioural markers. With log compression, the distance contribution of prior cancellations is comparable to that of channel and deposit type, enabling the algorithm to find the specific *combination* that defines C2.

Are the clusters coherent, stable, and interpretable? K-Means at $K = 5$ produces coherent clusters (silhouette 0.152, DB 1.701) with a clear profile narrative for each. Stability at $K = 5$ under StandardScaler is moderate (ARI 0.723 ± 0.270), driven by occasional seed-dependent label switching between C4 and adjacent groups. Under RobustScaler, stability is near-perfect (ARI 0.992 ± 0.003). The GMM confirms the hard cluster structure (assignment certainty ≈ 1.0 for most bookings) and the consensus matrix confirms that C2 is robustly identified across both modelling families.

What recommendations are supported? Based on post-hoc evidence:

1. **Targeted overbooking for C2:** Since 98.9% of C2 bookings historically cancel, hotels can confidently overbook this segment. This requires real-time classification at booking time using the pre-arrival features that define C2.
2. **Protect inventory for C3:** C3 (14.0% cancellation) represents last-minute direct bookings that almost always arrive; overbooking policies should account for this and avoid double-selling to C3 guests.
3. **Review deposit policy for C2:** The non-refundable deposit is already being applied to C2-type bookings, but forfeiture is not sufficient compensation for lost revenue from an $\approx 99\%$ cancellation cluster.

What should not be concluded? The 98.9% cancellation rate for C2 is a population-level *historical* estimate, not a deterministic prediction for any individual booking. Assigning any new booking to C2 based on its pre-arrival profile carries uncertainty. Furthermore, the clusters were formed from two specific Portuguese hotels over 2015–2017; generalisability to other properties or periods is not established. Nationality (captured via `country`)

was included to model geographic booking patterns, but should not be used to discriminate against individual guests from specific countries.

7 Limitations, Ethics, and Reproducibility

Limitations.

- **No ground-truth labels:** there are no external cluster labels against which to validate. Internal indices and stability measures provide relative, not absolute, quality guarantees.
- **Modest silhouette scores** (0.08–0.16 across the grid) reflect the genuine difficulty of partitioning a 55-dimensional mixed-type space. Euclidean distances in high-dimensional one-hot spaces are less informative than in low-dimensional numerical spaces.
- **Dataset scope:** two Portuguese hotels over three years. Results may not generalise.
- **GMM scalability:** full-covariance GMM in 55 dimensions requires substantial runtime (≈ 34 s per run per seed) and may be numerically sensitive; the high-dimensional covariance matrices may be poorly conditioned.
- **ADR filtering:** 4 records with $\text{ADR} < 0$ and extreme ADR outliers were removed as data-quality anomalies. Since ADR is excluded from the clustering matrix, this decision does not affect the R-EUCLID representation, but it should be documented for full reproducibility.

Ethics and governance. We follow the CC BY 4.0 terms of the dataset. The raw CSV is not committed to version control; `README.md` provides download and verification instructions. Nationality (`country`) is included as a behavioural proxy for geographic booking patterns, not as a basis for individual profiling or discrimination. Post-hoc cancellation rates are aggregate statistics; no individual guest should be penalised based solely on cluster membership.

Academic integrity. All code, analyses, and figures were produced by the team. Generative AI tools were consulted for clarifying methodological concepts and reviewing syntax; they were not used to generate substantive code, analyses, or report text.

Reproducibility.

- Repository tag: `Submission`
- Entry point: `python run_all.py` (full run) or `python run_all.py -fast` (10,000-row subsample, for pipeline integrity verification).
- Environment: `environment.yml` (Python 3.12, scikit-learn, pandas, numpy).
- Experiment log: `tables/experiments.csv` (324 rows; one per method/K/seed/representation combination).
- All figures and tables are generated automatically from `run_all.py` and the notebooks; no manual steps are required.

8 Conclusion

We conducted a reproducible, leakage-aware clustering study on 119,388 hotel booking records, constructing a 55-dimensional Euclidean representation from pre-arrival behavioural features. Applying K-Means ($K = 5$) and GMM as complementary families, with iK-Means for data-driven K estimation, we identified five structurally distinct booking profiles with post-hoc cancellation rates ranging from 5.5% to 98.9%.

The answer to RQ1 is affirmative: bookings made far in advance through indirect channels, combined with a high prior-cancellation history and non-refundable deposits, *do* naturally emerge as a structurally distinct cluster (C2; $n = 5,886$; 4.9% of bookings) without any outcome variable being used in the clustering step. The near-perfect historical cancellation rate of this cluster supports targeted overbooking and revised deposit policies as evidence-based operational recommendations.

The robustness analysis confirms that the core findings are stable across StandardScaler and RobustScaler and with/without the hotel binary feature. Future work should extend the analysis to additional hotels, validate the classification scheme on a holdout period, and explore whether a non-linear representation (e.g., UMAP embedding followed by clustering) would better separate the overlapping majority clusters.

References

- António, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41–49. <https://doi.org/10.1016/j.dib.2018.11.126>
- Mirkin, B. (2012). *Clustering: A Data Recovery Approach* (2nd ed.). Chapman and Hall/CRC.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.
- Caliński, T., & Harabasz, J. (1974). A dendrite method for cluster analysis. *Communications in Statistics*, 3(1), 1–27.
- Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1(2), 224–227.
- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218.
- Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1), 1–38.

Appendix: Figures

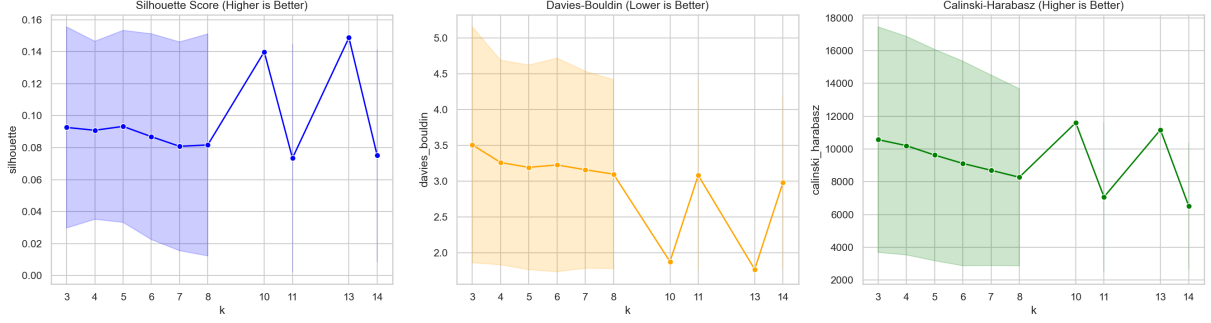


Figure 1: Internal validity indices (Silhouette, Davies-Bouldin, Calinski-Harabasz) for K-Means across $K \in \{3, \dots, 14\}$ on the main representation (R-EUCLID-standard). Shaded bands represent the range across 5 seeds. $K = 5$ is selected: CH is maximised (15,727) and DB is minimised among $K \leq 8$ (1.701), while the profile solution isolates the critical high-risk cluster.

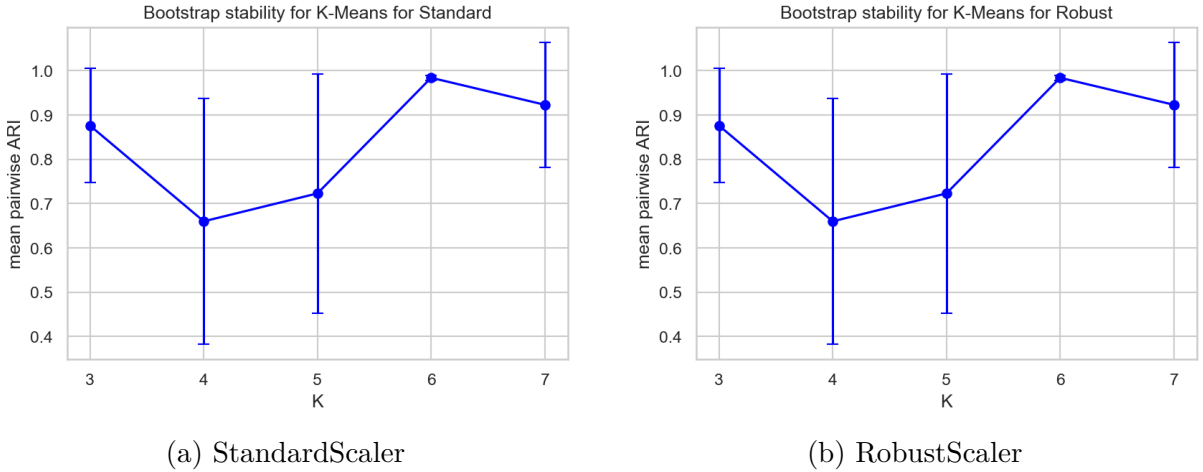


Figure 2: Bootstrap stability (mean pairwise ARI $\pm$ std, 190 resampling pairs) for K-Means across $K \in \{3, \dots, 7\}$. Both scalers show the same pattern shape; at $K = 5$ RobustScaler achieves ARI = 0.992 ± 0.003 (min 0.985) vs StandardScaler's 0.723 ± 0.270 (min 0.261), demonstrating that extreme values in `lead_time` and `previous_cancellations` can occasionally flip a seed under StandardScaler.

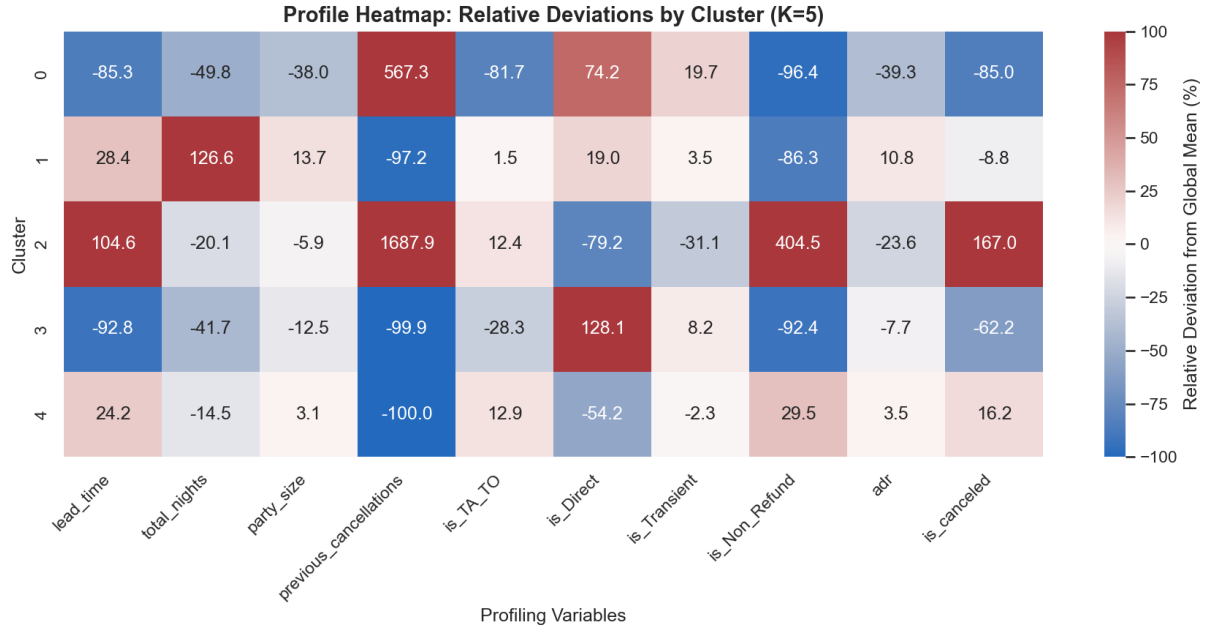


Figure 3: Profile heatmap: relative deviation from the global mean (%) for 10 key variables across 5 clusters (K-Means, $K = 5$). ADR and `is_canceled` are post-hoc profiling variables; all others are clustering inputs. Cluster 2 stands out for its extreme deviations in `previous_cancellations` (+1,688%) and `is_Non_Refund` (+405%), combined with the highest lead time (+105%) and an observed cancellation rate (+167% above mean = 98.9%).

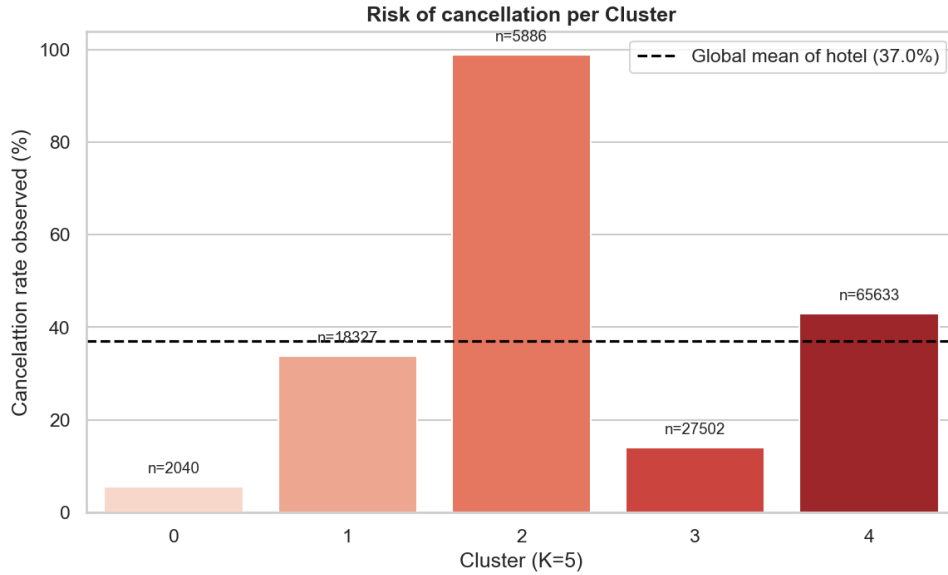
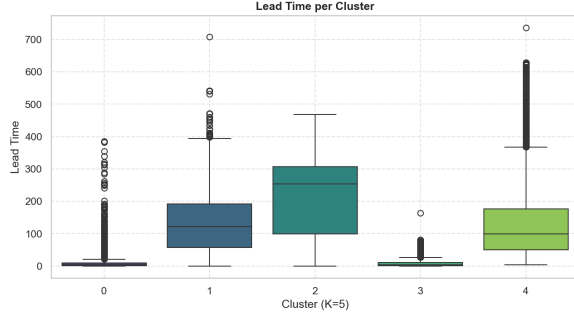
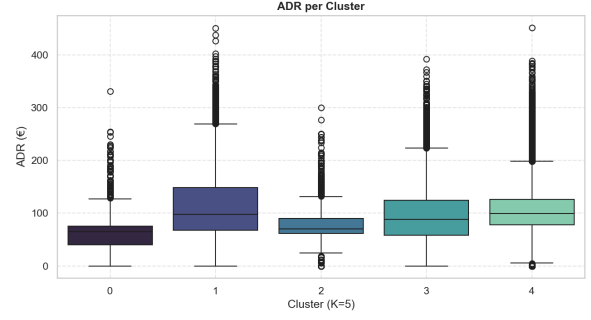


Figure 4: Post-hoc cancellation rate per cluster (K-Means, $K = 5$). The dashed line marks the global mean of 37.0%. Cluster 2 ($n = 5,886$) reaches 98.9%; Cluster 3 ($n = 27,502$) attains only 14.0%. Note that `is_canceled` was *not* used as a clustering input; these rates emerge purely from post-hoc profiling.



(a) Lead time distribution



(b) ADR distribution (post-hoc)

Figure 5: Lead time (clustering input) and ADR (post-hoc) box plots by cluster. Cluster 2 exhibits the longest lead times (median ≈ 250 days) and below-average ADR ($\approx \text{€}72$), consistent with low-cost speculative advance bookings. Cluster 3's near-zero lead time reflects last-minute direct bookings. ADR is only used for post-hoc profiling.

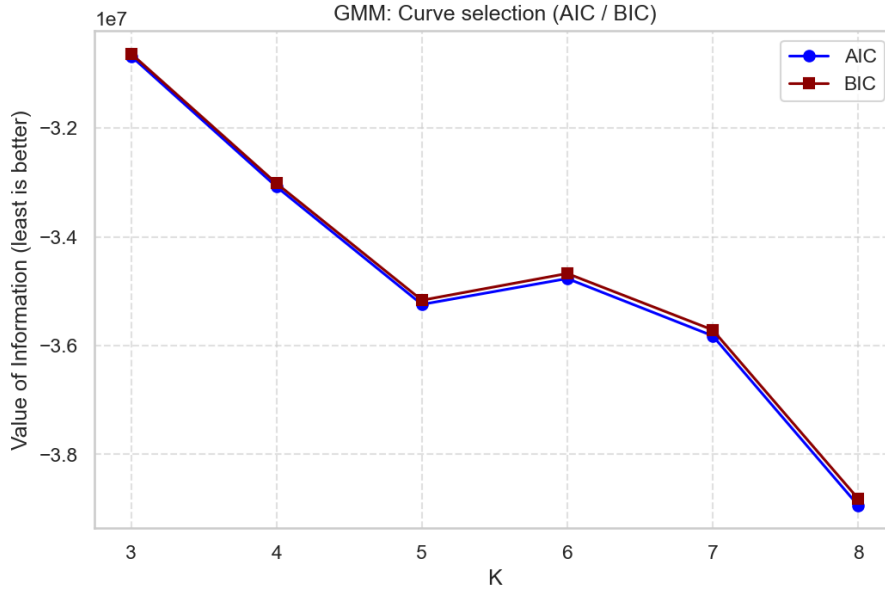


Figure 6: GMM model-selection curves (AIC and BIC) for $K \in \{3, \dots, 8\}$. Both criteria decrease monotonically without a clear elbow, indicating the data supports more components than the practical range tested. $K = 5$ is adopted for consistency with K-Means; a larger K may be justified if interpretability of more granular components is a priority.

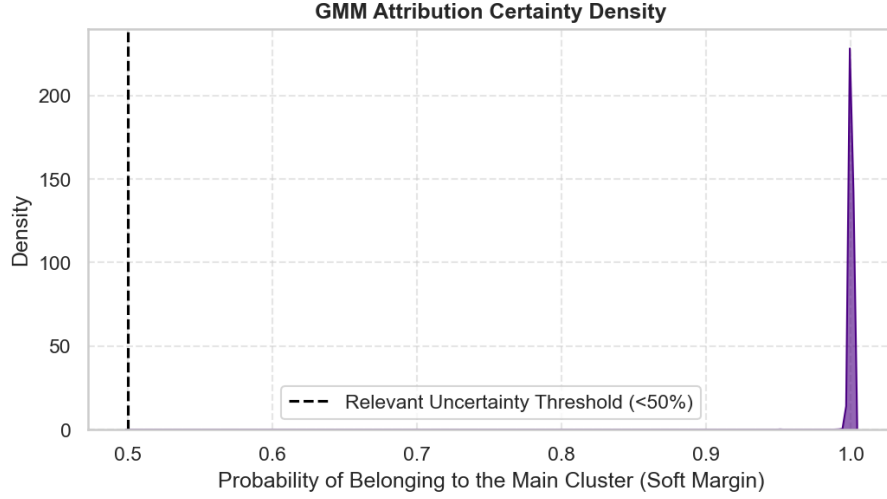


Figure 7: GMM assignment certainty density. The distribution is highly concentrated near $p = 1.0$, indicating that despite the soft probabilistic model, the great majority of bookings are unambiguously assigned to a single component. This supports the near-hard cluster structure found by K-Means.

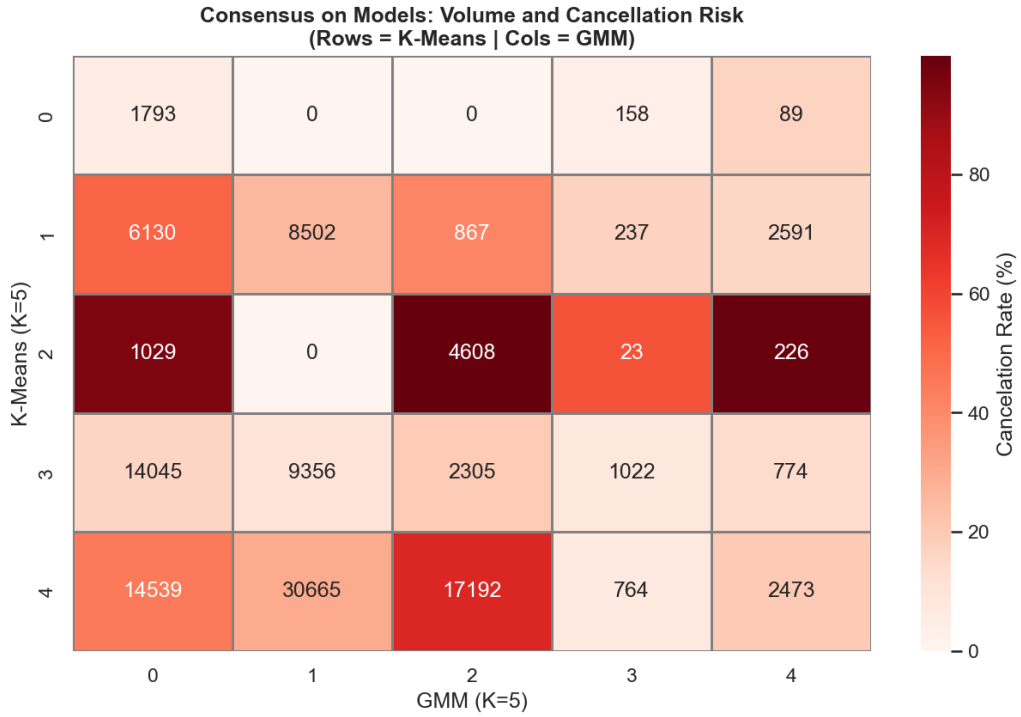


Figure 8: Consensus matrix between K-Means and GMM ($K = 5$). Rows are K-Means clusters; columns are GMM components; cell values are booking counts; colour encodes post-hoc cancellation rate. K-Means Cluster 2 (98.9% cancellation) maps almost entirely to GMM Cluster 2 (dark red diagonal cell: 4,608 bookings), confirming that the critical high-risk segment is robustly identified across both modelling families.

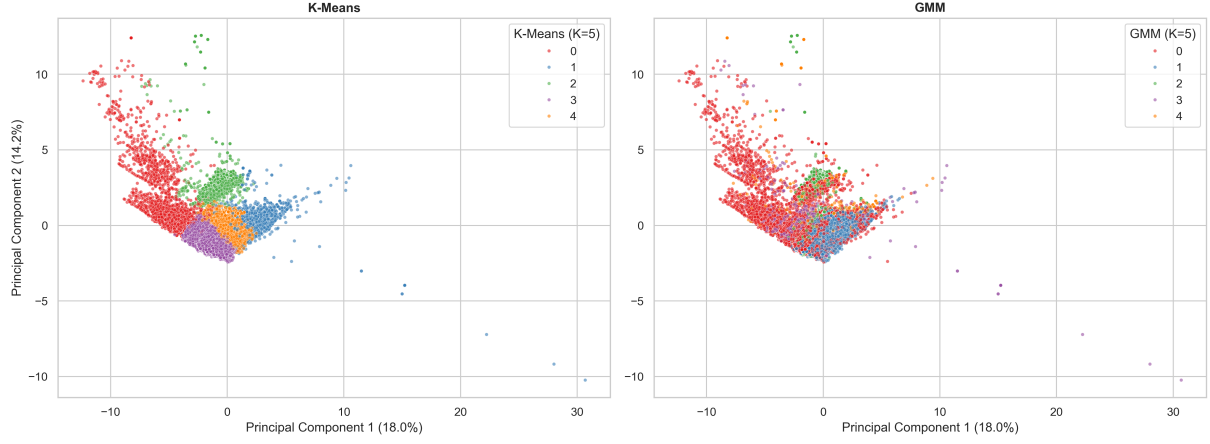


Figure 9: PCA projection (components 1 and 2, explaining 18.0% and 14.2% of variance respectively) of K-Means vs GMM assignments at $K = 5$. Both methods produce broadly similar spatial partitions. Cluster 2 (K-Means: green; GMM: green) occupies the upper-left region, corresponding to high lead time and high prior cancellation score in the PCA embedding. Note that PCA is used here for visualisation only; clustering was performed in the full 55-dimensional R-EUCLID space.